

Wasserstein Gradient Flows

Statements, hints and worked solutions

Each exercise is followed by question-by-question hints and a worked solution. Try the questions before reading the hints.

Exercise 1 (φ -divergences and Wasserstein gradient flow).

Let $\nu \in \mathcal{P}_2(\mathbb{R}^d)$ be a reference probability measure and let $\varphi : [0, \infty) \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex, lower semicontinuous and C^2 on $(0, \infty)$ with $\varphi(1) = 0$. For a probability measure α such that $\alpha \ll \nu$ with density $r := \frac{d\alpha}{d\nu}$, the φ -divergence of α with respect to ν is defined by

$$D_\varphi(\alpha \mid \nu) := \int_{\mathbb{R}^d} \varphi\left(\frac{d\alpha}{d\nu}(x)\right) d\nu(x) = \int_{\mathbb{R}^d} \varphi(r(x)) d\nu(x).$$

- (1) Show that $D_\varphi(\alpha \mid \nu) \geq 0$.
- (2) Consider the functional $f(\alpha) := D_\varphi(\alpha \mid \nu)$ on $\mathcal{P}_2(\mathbb{R}^d)$. Assume that both ν and α admit smooth strictly positive densities with respect to Lebesgue measure.
 - (a) Compute the first variation $\frac{\delta f}{\delta \alpha}(\alpha)(x)$ and deduce the Wasserstein gradient $\mathbb{W}f(\alpha)(x)$.
 - (b) Write the corresponding Wasserstein gradient flow PDE.
 - (c) Specialize your formulas to the case $\varphi(s) = s \log s$.

Useful hints

- (1) Apply Jensen with respect to the probability ν .
- (2a) Perturb the Lebesgue density while keeping the reference density fixed.
- (2b) The velocity is minus the spatial gradient of the first variation; insert it into the continuity equation.
- (2c) Use $\rho \nabla \log \rho = \nabla \rho$.

Detailed solution

(1) Since $r = d\alpha/d\nu$ has $\int r d\nu = 1$, Jensen gives $D_\varphi(\alpha \mid \nu) \geq \varphi(\int r d\nu) = \varphi(1) = 0$. If φ is strictly convex, equality holds exactly when $r = 1$ ν -a.e., that is, $\alpha = \nu$.

(2a) Write $d\alpha = \rho dx$ and $d\nu = n dx$, so the relative density is $r = \rho/n$. For a smooth compactly supported perturbation η with $\int \eta = 0$, differentiation gives

$$\left. \frac{d}{dh} f((\rho + h\eta) dx) \right|_{h=0} = \int \varphi'(\rho/n) \eta dx.$$

Consequently, up to an additive constant,

$$\frac{\delta f}{\delta \alpha} = \varphi'(\rho/n), \quad \mathbb{W}f(\alpha) = \nabla[\varphi'(\rho/n)].$$

The double nabla denotes the Wasserstein gradient. The PDE calculations here and below assume sufficient smoothness and integrability; they identify the gradient field, not a general PDE existence theorem.

(2b) With $v_t = -\mathbb{W}f(\alpha_t)$, the continuity equation becomes

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla[\varphi'(\rho_t/n)]).$$

Under vanishing boundary flux, it conserves mass and satisfies $\frac{d}{dt} f(\alpha_t) = - \int |\nabla f(\alpha_t)|^2 \rho_t dx \leq 0$.

(2c) For $\varphi(s) = s \log s$, the additive 1 in $\varphi'(s)$ disappears under the spatial gradient. Thus

$$\nabla f(\alpha) = \nabla \log(\rho/n), \quad \partial_t \rho_t = \Delta \rho_t - \nabla \cdot (\rho_t \nabla \log n).$$

If $n = Z^{-1} e^{-V}$, this is $\partial_t \rho_t = \Delta \rho_t + \nabla \cdot (\rho_t \nabla V)$; the standard Gaussian corresponds to $V(x) = |x|^2/2$. For comparison, reverse KL uses $\varphi(s) = -\log s$ and gives $\partial_t \rho_t = -\nabla \cdot (\rho_t \nabla (n/\rho_t))$.

Exercise 2 (Gradient flows of Dirichlet energy and Fisher information).

To avoid boundary terms, work on the flat unit-volume torus $\mathbb{T}^d$, with a smooth strictly positive density ρ and $d\alpha = \rho dx$. The same local differential expressions hold on $\mathbb{R}^d$ when the required integrations by parts are justified.

(1) For the Dirichlet energy $\mathcal{E}(\alpha) = \int |\nabla \rho|^2 dx$:

- (a) Compute the first variation and the unconstrained $L^2(dx)$ descent equation $\partial_t \rho = -\delta \mathcal{E} / \delta \rho$.
- (b) Compute the Wasserstein gradient and its continuity-equation PDE.

(2) Repeat these computations for the Fisher information

$$\mathcal{I}(\alpha) = \int \rho |\nabla \log \rho|^2 dx = \int \frac{|\nabla \rho|^2}{\rho} dx.$$

Does its unconstrained L^2 flow preserve mass? How should it be modified to preserve $\int \rho = 1$? All evolution equations are to be understood while the solution is smooth and positive.

Useful hints

- (1a) Differentiate $\int |\nabla(\rho + h\eta)|^2$ and integrate by parts.
- (1b) A Wasserstein velocity is minus the spatial gradient of the first variation, not minus the first variation itself.
- (2) Differentiate both the numerator and denominator of $|\nabla \rho|^2 / \rho$. For the mass constraint, project the L^2 gradient onto zero-mean functions.

Detailed solution

(1a) Differentiating at $h = 0$ gives

$$\left. \frac{d}{dh} \mathcal{E}((\rho + h\eta) dx) \right|_{h=0} = 2 \int \nabla \rho \cdot \nabla \eta dx = -2 \int \Delta \rho \eta dx.$$

Hence $\delta \mathcal{E} / \delta \rho = -2\Delta \rho$ and the L^2 descent equation is $\partial_t \rho = 2\Delta \rho$. The sign is negative-gradient descent; this is the heat equation, which preserves mass.

(1b) The Wasserstein gradient is $-2\nabla \Delta \rho$, so its velocity is $2\nabla \Delta \rho$. Therefore

$$\partial_t \rho = -2\nabla \cdot (\rho \nabla \Delta \rho).$$

This fourth-order equation is different from L^2 descent of the same energy. Linearization about a constant density $\bar{\rho}$ yields $\partial_t \eta = -2\bar{\rho} \Delta^2 \eta$.

(2) The density factor in Fisher information is essential. Differentiating gives

$$\left. \frac{d}{dh} \mathcal{I}((\rho + h\eta) dx) \right|_{h=0} = \int \left(\frac{2\nabla \rho \cdot \nabla \eta}{\rho} - \frac{|\nabla \rho|^2}{\rho^2} \eta \right) dx.$$

Integrating the first term by parts yields

$$q := \frac{\delta \mathcal{I}}{\delta \rho} = -2\Delta \log \rho - |\nabla \log \rho|^2 = -4 \frac{\Delta \sqrt{\rho}}{\sqrt{\rho}}.$$

Consequently the unconstrained L^2 equation is $\partial_t \rho = -q = 2\Delta \log \rho + |\nabla \log \rho|^2$. Its mass derivative is $\int |\nabla \log \rho|^2 dx$, generally positive. On the unit-volume torus the mass-constrained L^2 equation is instead

$$\partial_t \rho = -q + \int q dx = 2\Delta \log \rho + |\nabla \log \rho|^2 - \int |\nabla \log \rho|^2 dx.$$

This projection enforces mass, but by itself is not a proof that positivity is preserved.

For the Wasserstein geometry, $\mathbb{W}I = \nabla q$, giving

$$\partial_t \rho = \nabla \cdot (\rho \nabla q) = -\nabla \cdot \left(\rho \nabla [2\Delta \log \rho + |\nabla \log \rho|^2] \right).$$

It conserves mass by its divergence form and dissipates $\mathcal{I}$ at rate $-\int \rho |\nabla q|^2 dx$. The unweighted integral $\int |\nabla \log \rho|^2 dx$ would be a different functional, not Fisher information.

Exercise 3 (Gaussian evolution under linear dynamics).

Let $(X^{(\ell)})_{\ell \in \mathbb{N}}$ be a sequence of random variables in $\mathbb{R}^d$ defined by the linear recursion

$$X^{(\ell+1)} = X^{(\ell)} - \tau A^{(\ell)} X^{(\ell)}, \quad \ell = 0, 1, 2, \dots,$$

where $\tau > 0$ is a fixed step size and, for each ℓ , $A^{(\ell)} \in \mathbb{R}^{d \times d}$ is a deterministic matrix. Denote by $\alpha^{(\ell)}$ the law of $X^{(\ell)}$.

- (1) Assume that the initial condition is Gaussian $X^{(0)} \sim \mathcal{N}(0, \Sigma^{(0)})$, for some symmetric positive semidefinite covariance matrix $\Sigma^{(0)} \in \mathbb{R}^{d \times d}$. Show that each $X^{(\ell)}$ is also Gaussian, and describe the evolution of the law $\alpha^{(\ell)} = \text{Law}(X^{(\ell)})$ by giving an explicit recursion for the covariance matrices $\Sigma^{(\ell)}$.
- (2) Consider now the continuous time ODE $\dot{X}_t = -A_t X_t$, $t \geq 0$, with $A_t \in \mathbb{R}^{d \times d}$ a (say, continuous) time dependent matrix and random initial condition $X_0 \sim \mathcal{N}(0, \Sigma_0)$. Explain why X_t is Gaussian for every $t \geq 0$, say $X_t \sim \mathcal{N}(0, \Sigma_t)$, and, by considering the limit $\tau \rightarrow 0$ in the discrete time recursion from part (1), show that $(\Sigma_t)_{t \geq 0}$ solves a matrix ODE that you should identify.

Useful hints

- (1) A deterministic linear image of a Gaussian is Gaussian. Compute the covariance of $(I - \tau A)X$.
- (2) Introduce the fundamental matrix, differentiate $\Phi_t \Sigma_0 \Phi_t^\top$, and compare with the first-order expansion of the discrete recursion.

Detailed solution

- (1) Write $B^{(\ell)} = I - \tau A^{(\ell)}$. If $X^{(\ell)} \sim \mathcal{N}(0, \Sigma^{(\ell)})$, then its linear image is Gaussian and

$$\Sigma^{(\ell+1)} = B^{(\ell)} \Sigma^{(\ell)} (B^{(\ell)})^\top.$$

Induction proves the claim, including singular covariances. No stability bound on τ is needed just for Gaussian preservation; it would matter for long-time numerical stability.

- (2) Let $\dot{\Phi}_t = -A_t \Phi_t$, $\Phi_0 = I$. The solution is $X_t = \Phi_t X_0$, so $\alpha_t = \mathcal{N}(0, \Phi_t \Sigma_0 \Phi_t^\top)$. Differentiating gives the rigorous covariance ODE

$$\dot{\Sigma}_t = -A_t \Sigma_t - \Sigma_t A_t^\top.$$

To compare with the discrete scheme, take $A^{(\ell)} = A_{\ell\tau}$ and expand

$$(I - \tau A)\Sigma(I - \tau A)^\top = \Sigma - \tau(A\Sigma + \Sigma A^\top) + \tau^2 A\Sigma A^\top.$$

Thus the difference quotient is the displayed vector field plus an $O(\tau)$ term on bounded sets. The fundamental-matrix argument proves the ODE independently of this consistency calculation.

Exercise 4 (Wasserstein flow for quadratic kernels).

Let $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ and denote its mean and covariance by

$$m(\alpha) := \int_{\mathbb{R}^d} x d\alpha(x), \quad \Sigma(\alpha) := \int_{\mathbb{R}^d} (x - m(\alpha))(x - m(\alpha))^\top d\alpha(x).$$

Consider the functionals

$$f(\alpha) := \iint_{\mathbb{R}^d \times \mathbb{R}^d} k(x, y) d\alpha(x) d\alpha(y)$$

on the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$, where the kernels k are given below.

- (1) Let $k(x, y) = \langle Ax, y \rangle$, $A \in \mathbb{R}^{d \times d}$.
 - (a) Express $f(\alpha)$ in terms of $m(\alpha)$ and A .
 - (b) Compute the Wasserstein gradient $\mathbb{W}f(\alpha)$.
 - (c) Write the Wasserstein gradient flow PDE, and, for a general initial condition $\alpha_{t=0} = \alpha_0$, give an explicit expression of α_t in terms of α_0 .
- (2) Let $k(x, y) = \langle Ax, y \rangle \langle Bx, y \rangle$, $A, B \in \mathbb{R}^{d \times d}$.
 - (a) Compute the Wasserstein gradient $\mathbb{W}f(\alpha)$.
 - (b) Using the previous exercise on linear ODEs with Gaussian initial data, show that if $\alpha_{t=0} = \mathcal{N}(0, \Sigma_0)$, then $\alpha_t = \mathcal{N}(0, \Sigma_t)$ for as long as the covariance ODE exists, and derive the matrix ODE satisfied by $(\Sigma_t)_{t \geq 0}$.

Useful hints

- (1a) Integrate the bilinear kernel in each argument.
- (1b) Only $(A + A^\top)/2$ contributes to a quadratic form.
- (1c) A spatially constant velocity translates the law; first solve the mean ODE.
- (2a) Differentiate both copies of α . Use the raw second moment $S = \int x x^\top d\alpha$, not just the covariance.
- (2b) Apply the preceding covariance equation to the resulting linear field. Arbitrary A, B need not yield a global solution.

Detailed solution

- (1a) Independence in the double integral gives $f(\alpha) = m^\top A m = m^\top A_s m$, where $A_s = (A + A^\top)/2$.
- (1b) For a zero-mass variation η , $\dot{m} = \int x d\eta(x)$, so $Df(\alpha)[\eta] = \int 2\langle A_s m, x \rangle d\eta(x)$. Therefore

$$\frac{\delta f}{\delta \alpha}(x) = 2\langle A_s m, x \rangle, \quad \mathbb{W}f(\alpha)(x) = 2A_s m.$$

- (1c) The velocity is constant in space: $v_t = -2A_s m_t$. Integrating the continuity equation against x gives $\dot{m}_t = -2A_s m_t$, hence $m_t = e^{-2A_s t} m_0$. Consequently

$$\alpha_t = (x \mapsto x + m_t - m_0) \# \alpha_0.$$

This solves $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ for any initial $\alpha_0 \in \mathcal{P}_2$, including singular laws. Its covariance is constant. No inverse of A_s is required.

(2a) Put $S = \int x x^\top d\alpha = \Sigma + m m^\top$. Varying both factors of the product measure gives

$$\frac{\delta f}{\delta \alpha}(x) = \int [k(x, y) + k(y, x)] d\alpha(y) = x^\top (A^\top S B + A S B^\top) x.$$

For a matrix D , $\nabla(x^\top D x) = (D + D^\top)x$. Thus

$$\mathbb{W}f(\alpha)(x) = M(S)x, \quad M(S) = A^\top S B + B^\top S A + A S B^\top + B S A^\top.$$

This formula holds for arbitrary, not necessarily symmetric, A, B . Both differentiations are needed because k need not be symmetric. Replacing S by Σ without assuming $m = 0$ would be incorrect.

(2b) At zero mean, $S = \Sigma$. Solve the locally Lipschitz matrix ODE

$$\dot{\Sigma}_t = -M(\Sigma_t)\Sigma_t - \Sigma_t M(\Sigma_t), \quad \Sigma_{t=0} = \Sigma_0.$$

Along this deterministic covariance path the characteristic equation is linear, $\dot{X}_t = -M(\Sigma_t)X_t$. Its fundamental matrix therefore sends the initial Gaussian to a centered Gaussian with exactly the covariance above. This constructs the claimed Wasserstein flow on the interval of existence.

There is no general assertion of global existence for arbitrary A, B : in dimension one, take $A = 1$, $B = -1$ and variance $s_0 > 0$. Then $f = -s^2$, $M(s) = -4s$, and $\dot{s} = 8s^2$, giving $s_t = s_0/(1 - 8s_0 t)$ and finite-time blow-up.

Further exercises

Exercise 5 (Mean-field two-layer linear network under Wasserstein flow).

Let $\rho \in \mathcal{P}_2(\mathbb{R}^d)$ be the data distribution and let $h \in L^2(\rho)$ be a target function. For a parameter distribution

$$\alpha \in \mathcal{P}_2(\mathbb{R} \times \mathbb{R}^d), \quad x = (u, v) \in \mathbb{R} \times \mathbb{R}^d,$$

consider the two-layer linear mean-field model

$$g_\alpha(z) := \int u \langle z, v \rangle d\alpha(u, v), \quad z \in \mathbb{R}^d,$$

and the quadratic loss

$$f(\alpha) := \int_{\mathbb{R}^d} |g_\alpha(z) - h(z)|^2 d\rho(z).$$

(1) Show that f can be written as a quadratic functional of α of the form

$$f(\alpha) = \iint k(x, y) d\alpha(x) d\alpha(y) + \int b(x) d\alpha(x) + c,$$

where $x = (u, v)$, $y = (u', v')$, and identify explicit formulas for $k(x, y)$ and $b(x)$ in terms of ρ and h . You may leave the constant c implicit.

(2) Consider the Wasserstein gradient flow of f on $\mathcal{P}_2(\mathbb{R} \times \mathbb{R}^d)$:

$$\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0, \quad v_t(x) = -\mathbb{W}f(\alpha_t)(x).$$

Using the structure from part (1) and the previous exercise on quadratic kernels, show that if

$$\alpha_{t=0} = \mathcal{N}(0, \Sigma_0),$$

then α_t remains Gaussian for all $t \geq 0$.

Useful hints

- (1) Introduce $K = \int zz^\top d\rho$ and $c_h = \int zh(z) d\rho$. Expand the square.
- (2) The predictor depends on $q = \int uv d\alpha$. Compute the parameter gradient explicitly, then use energy dissipation to bound the linear coefficient and exclude finite-time blow-up.

Detailed solution

(1) Define the raw second-moment matrix $K = \int zz^\top d\rho(z)$ and $c_h = \int zh(z) d\rho(z)$. They are finite by the stated moment assumptions and Cauchy–Schwarz. Put $q = \int uv d\alpha(u, v)$; this is finite because $|u||v| \leq (u^2 + |v|^2)/2$. Then

$$g_\alpha(z) = z^\top q, \quad f(\alpha) = q^\top K q - 2c_h^\top q + \int h^2 d\rho.$$

Expanding q yields the requested kernel and linear term:

$$k((u, v), (u', v')) = uu'v^\top Kv', \quad b(u, v) = -2u c_h^\top v.$$

The kernel is quadratic in each parameter vector separately, not bilinear in those vectors. Also, K is a covariance only if the data have zero mean.

(2) Let $r = Kq - c_h$. The first variation is $2u r^\top v$ up to a constant, hence

$$\mathbb{W}f(\alpha)(u, v) = \begin{pmatrix} 2r^\top v \\ 2ur \end{pmatrix} = M(\alpha) \begin{pmatrix} u \\ v \end{pmatrix}, \quad M(\alpha) = 2 \begin{pmatrix} 0 & r^\top \\ r & 0 \end{pmatrix}.$$

There is no affine constant term. At zero mean, $q = \Sigma_{v,u}$ is the cross-covariance, so M depends only on Σ and

$$\dot{\Sigma}_t = -M(\Sigma_t)\Sigma_t - \Sigma_t M(\Sigma_t), \quad \alpha_t = \mathcal{N}(0, \Sigma_t).$$

This initially gives Gaussian closure on the local existence interval. To prove it lasts for all finite times, use

$$|r_t| = \left| \int z(g_{\alpha_t}(z) - h(z)) d\rho(z) \right| \leq \left(\int |z|^2 d\rho \right)^{1/2} \sqrt{f(\alpha_t)}.$$

Along the smooth moment flow, $\frac{d}{dt}f(\alpha_t) = -\int |\mathbb{W}f(\alpha_t)|^2 d\alpha_t \leq 0$. Thus r_t , and hence $\|M(\Sigma_t)\|$, are bounded by constants depending only on the initial loss and the data. Gronwall bounds $\text{tr} \Sigma_t$ on every finite interval, preventing finite-time escape of the covariance ODE. The Gaussian flow is therefore global and remains centered.

Exercise 6 (Wasserstein flow of W_2^2).

Fix $\beta \in \mathcal{P}_2(\mathbb{R}^d)$. For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ define $f(\alpha) := W_2(\alpha, \beta)^2$. For the first-variation calculation, assume α has a smooth strictly positive density and that the usual envelope differentiation is justified for the perturbations considered. For the flow take an absolutely continuous initial law $\alpha_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Consider the quadratic cost $c(x, y) = \frac{1}{2}|x - y|^2$.

- (1) Using the Kantorovich dual problem, and using the envelope theorem (that was used the same way in the previous exercise sheet for discrete measure), compute the first variation $\delta f(\alpha)$.
- (2) Assume α is absolutely continuous. Leveraging Brenier's theorem connecting the transport map T to the dual potential, compute the Wasserstein gradient $\mathbb{W}f(\alpha)$.
- (3) Relate the Wasserstein flow $(\alpha_t)_{t \geq 0}$ to the McCann interpolation (you should use Benamou–Brenier dynamical formulation which details a valid field advecting the interpolant).

Useful hints

- (1) The dual for $|x - y|^2/2$ represents $f/2$, not f .
- (2) Write the source dual potential as $|x|^2/2 - u(x)$, with $T = \nabla u$.
- (3) If $I_s = (1 - s)\text{Id} + sT$, compute $\frac{d}{ds}I_s$. At the current point, the remaining displacement to the target is $(1 - s)$ times this velocity.

Detailed solution

(1) Let φ_α be a source optimal potential for cost $|x - y|^2/2$. Kantorovich duality represents $f(\alpha)/2$ as the supremum of $\int \varphi d\alpha + \int \psi d\beta$ over $\varphi + \psi \leq |x - y|^2/2$. Under the differentiability hypothesis, its derivative in a zero-mass direction η is $\int \varphi_\alpha d\eta$. Thus

$$\frac{\delta f}{\delta \alpha}(x) = 2\varphi_\alpha(x) + \text{constant}.$$

Even without differentiability, the optimal potential gives the corresponding subgradient inequality in the linear space of measures. This should not be confused with geodesic convexity.

(2) Brenier's theorem gives $T_\alpha = \nabla u_\alpha$ with u_α convex, and $\varphi_\alpha(x) = |x|^2/2 - u_\alpha(x)$ up to a constant. Therefore

$$\mathbb{W}f(\alpha)(x) = 2(x - T_\alpha(x)), \quad v(x) = 2(T_\alpha(x) - x).$$

The factor 2 comes from the definition $f = \mathcal{W}_2^2$, rather than $f = \mathcal{W}_2/2$.

(3) Let T be the optimal map from α_0 to β , put $I_s = (1 - s)\text{Id} + sT$, and write $\mu_s = (I_s)_\# \alpha_0$. For $s < 1$, monotonicity of T makes I_s strongly monotone and its inverse on its image Lipschitz. In particular μ_s is absolutely continuous. The remaining optimal map is $T_s = T \circ I_s^{-1}$; the optimality follows from the constant-speed geodesic property of displacement interpolation.

At $x = I_s(x_0)$, the geodesic velocity is

$$w_s(x) = T(x_0) - x_0 = \frac{T_s(x) - x}{1 - s}.$$

It is *not* merely $T_s(x) - x$. This field solves $\partial_s \mu_s + \nabla \cdot (\mu_s w_s) = 0$ and has norm $\|w_s\|_{L^2(\mu_s)} = \mathcal{W}_2(\alpha_0, \beta)$, as in Benamou–Brenier.

For $\alpha_t = \mu_{s(t)}$, its velocity is $s'(t)w_{s(t)}$. Matching it with the negative gradient $2(T_{s(t)} - \text{Id}) = 2(1 - s(t))w_{s(t)}$ gives

$$s'(t) = 2(1 - s(t)), \quad s(0) = 0, \quad \alpha_t = (e^{-2t}\text{Id} + (1 - e^{-2t})T)_\# \alpha_0.$$

Consequently $\mathcal{W}_2(\alpha_t, \beta) = e^{-2t}\mathcal{W}_2(\alpha_0, \beta)$ and $f(\alpha_t) = e^{-4t}f(\alpha_0)$. Moreover $|\dot{\alpha}_t| = 2\mathcal{W}_2(\alpha_t, \beta) = \|\mathbb{W}f(\alpha_t)\|_{L^2(\alpha_t)}$, so this is a descending gradient-flow curve, with energy dissipation $f' = -\|\mathbb{W}f\|^2$. The endpoint is reached only as $t \rightarrow \infty$, unless $\alpha_0 = \beta$ already.